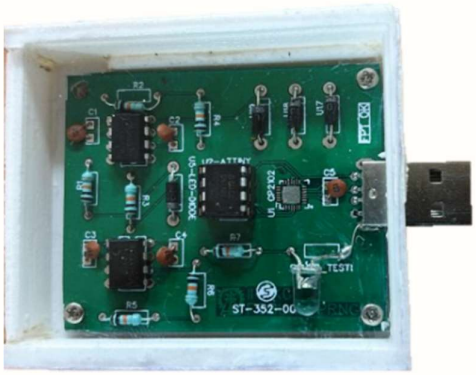
FONQP

RANDOM NUMBER GENERATOR

Indian Patent Filed (202431022600)

FONQP Random Number Generator is a True Hardware RNG design producing high-quality random bits using chaotic noise. This design has been packaged into a pluggable USB 2.0 module, making it compact and easy to use across various applications. The pipeline works by sampling an autonomous chaotic circuit to generate noise, followed by applying a stream conditioner to enhance randomness. The design is easily manufacturable with a high quality/cost ratio.



Specifications

min-Entropy (source)	**> 0.89 bits/bit**
min-Entropy (conditioned)	**> 0.98 bits/bit**
Device bit-rate	**~ 1Mbps (USB 2.0)**
Package dimension	**70 (l) x 35 (b) x 20 (h) mm^3**
Type	**Pluggable USB 2.0**

Compliance and Assessments

- **NIST SP 800-22 Statistical Test Suite** **Passed** ☑
- **Dieharder Test Suite** **Passed** ☑

Areas of Application

- ✓ **Internet of Things (IoT)**
- ✓ **Cryptographic Key Generation**
- ✓ **Classical and Quantum Communication**
- ✓ **Banking and Trading Sector**

Software and Driver Support: https://github.com/FONQP/fonqp-rng

In Linux environments, the driver registers the RNG as a ***hwrng*** device, automatically mixing entropy into the input pool. Therefore the numbers can be accessed using the ***/dev/random*** or ***getrandom()*** interface.

Other Resources: https://fonqp.github.io/#/patents/202431022600

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